Om Patel

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Education

Rutgers University–New Brunswick Sep 2025 – May 2027
Master of Science in Computer Science; GPA: 3.92/4.0 *New Brunswick, NJ*

Pandit Deendayal Energy University Jun 2021 – Jun 2025
Bachelor of Technology in Information & Communication Technology; CGPA: 8.93/10 *Gandhinagar, India*

Technical Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, R, HTML/CSS
Frameworks: React, Next.js, Node.js, FastAPI, Flask, Django, SQLAlchemy, Prisma, REST APIs, Streamlit
Databases: PostgreSQL, Redis, MySQL, SQLite, MongoDB
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, GitHub Actions (CI/CD), Vercel, Render, Git, Linux, Bash
Testing: pytest, Locust (load testing), unit and end-to-end testing
Machine Learning & Data: PyTorch, TensorFlow, Scikit-learn, LLM APIs, Pandas, NumPy, Qualtrics, Excel

Experience

Research & Assessment Graduate Coordinator (Part-Time) Jul 2026 – Present
Rutgers University – Career Exploration and Success *New Brunswick, NJ*

- Develop and maintain an employer-facing web dashboard that visualizes career fair attendance by major to attract new recruiters (Python, HTML/CSS, JSON); used by 10+ staff ahead of public launch
- Designed a rule-based mentor–mentee matching algorithm that scores student and mentor profiles, producing ∼40 pairs
- Map student engagement with Career Exploration and Success across majors in Python to target outreach by major

Research Intern – Deep Learning Jan 2025 – May 2025
Pandit Deendayal Energy University *Gandhinagar, India*

- Automated preprocessing pipelines for 10,000+ geospatial images (TensorFlow, PyTorch), cutting latency by 40%
- Engineered a Dual-Stream U-Net to detect temporal changes in satellite imagery, achieving 89% IoU and improving deforestation-detection F1 score by 17% across diverse test regions
- Outperformed FCN-8 baselines on land-cover reconstruction with optimized U-Nets, raising PSNR by 22% (0.85 SSIM)

Embedded Systems Intern May 2024 – Jul 2024
Candore Bioscience *Ahmedabad, India*

- Wrote firmware for a sensor-based inference algorithm that estimates cryogenic liquid levels from noisy thermal and ultrasonic data (ESP32 MicroPython, Arduino Embedded C), improving reliability by 40%
- Applied Steinhart-Hart regression and exponential smoothing to temperature signals, enabling stable readings to −23°C
- Programmed real-time data acquisition for sensor calibration, signal filtering, and anomaly-resistant measurement

Projects

EventPass | *Python, FastAPI, PostgreSQL, Redis, Docker, Locust* github.com/Ompatel179/Eventpass

- Built a QR-code event check-in platform for student clubs: FastAPI and PostgreSQL backend (SQLAlchemy, Alembic), Redis rate limiting, JWT auth with Argon2 hashing and role-based access, plus an organizer dashboard and phone scanner
- Guaranteed exactly-once check-in with a database unique constraint: 0 duplicates in 2,862 concurrent scans
- Diagnosed a deadlock under Locust load tests that froze the API at 13 req/s; after the fix it served ∼950 req/s with 0 errors on one worker, and one index cut a 1M-row query from 16.2 ms to 0.08 ms (82 tests, 97% coverage in CI)

Composur | *Next.js, TypeScript, PostgreSQL, Prisma, AWS S3, Vercel* composur.tech

- Architected the layered backend of an AI mock-interview platform (API → Services → Repositories → Prisma/PostgreSQL) with session auth, S3 presigned uploads, and Postgres-backed rate limiting
- Shipped to Vercel behind GitHub Actions CI running lint, type-checking, and unit and end-to-end tests on every push
- Built a real-time voice interview pipeline chaining speech recognition, streaming LLM responses, and low-latency text-to-speech (Llama 3.3 70B via Groq, ElevenLabs), with role-specific prompts for 28 interview domains

SRGAN Image Super-Resolution | *Python, PyTorch, GANs, VGG19, OpenCV*

- Implemented a Super-Resolution GAN (generator–discriminator) in PyTorch for 4× upscaling, scored with PSNR and LPIPS
- Created paired low-/high-resolution data loaders with augmentation, checkpointing, and inference workflows
- Trained the model in two stages (MSE pre-training, then adversarial fine-tuning with VGG19 perceptual loss)

Publications

Harnessing Generative Adversarial Models to Enhance Satellite Imagery Quality Aug 2025

- Published at the 2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA): SRGAN/ESRGAN framework for satellite image reconstruction achieving 34.55 dB PSNR and 0.08 LPIPS